Tutorial 13: EKF for Quadruped State Estimation

Athletic Intelligence in Robotics (AIR) Course

Mihaela Popescu

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Outline

Tutorial Goal

- Implement an EKF for quadruped state estimation.
- Fuse:
  - IMU measurements for prediction.
  - Stance-foot contacts for correction.
- Study the effect of:
  - IMU noise
  - IMU bias
  - foot slip
  - process and measurement noise tuning
  - missing contacts / flight phases

Main idea

IMU integration predicts motion, while stance feet provide temporary world-frame anchors that reduce drift.

State Representation in the EKF

- The EKF maintains an estimate of the system state:

$$\hat{x} = \left[R_{WB},\ v_W,\ p_W,\ p_{F,W}^{(1)},\ p_{F,W}^{(2)},\ \ldots \right]$$

  - $R_{WB} \in \mathrm{SO}(3)$: rotation from body frame $\{B\}$ to world frame $\{W\}$.
  - v_W: base velocity in world frame.
  - p_W: base position in world frame.
  - $p_{F,W}^{(i)}$: world-frame position of foot i during stance.

- The covariance is defined over an **error (perturbation) state** δx:

$$\delta x = \left[\delta\theta,\ \delta v,\ \delta p,\ \delta p_F^{(1)},\ \delta p_F^{(2)},\ \ldots \right]$$

- **Error-state EKF**: the filter estimates a small correction δx and injects it into the state, ensuring consistent handling of orientation on $\mathrm{SO}(3)$.

Contact-Dependent Foot States

- Foot states are added only when a foot enters contact.
- When a foot is in stance, the no-slip assumption gives:

$$p_{F,W} \approx \text{constant}$$

- When a new stance foot is detected, initialize its world position as:

$$p_{F,W} = p_W + R_{WB} p_{F,B}$$

- Here:
  - $p_{F,B}$ is the measured foot position in the body frame.
  - $p_{F,W}$ is the stored foot anchor in the world frame.
- When contact ends, the corresponding foot state is removed.

Note

A swing foot should not be treated as fixed, because it moves relative to the world.

Useful Operator: Skew Matrix

- The skew matrix represents the cross product:

$$[a]_\times b = a \times b$$

- For

$$a = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix},$$

the skew matrix is:

$$[a]_\times = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix}$$

- Small rotations are applied using the **matrix exponential**:

$$R_{\text{new}} = R \exp([\delta\theta]_\times)$$

Prediction Step: Orientation

- The **gyroscope** measures angular velocity in the **body frame**:

$$\omega_B = \begin{bmatrix} \omega_x & \omega_y & \omega_z \end{bmatrix}^T$$

- Over one time step Δt, compute:

$$\Delta R = \exp([\omega_B]_\times \Delta t)$$

- Propagate orientation:

$$R_{WB,k} = R_{WB,k-1} \Delta R$$

Prediction Step: Acceleration, Velocity, Position

- The **accelerometer** measures specific force in the **body frame**.
- Rotate it into the world frame and add gravity:

$$a_W = R_{WB,k} a_B + g_W$$

- Propagate velocity:

$$v_{W,k} = v_{W,k-1} + a_W \Delta t$$

- Propagate position:

$$p_{W,k} = p_{W,k-1} + v_{W,k-1} \Delta t + \frac{1}{2} a_W \Delta t^2$$

Prediction Step: Error-State Jacobian

- Covariance propagation:

$$P_k^- = F_k P_{k-1} F_k^T + Q$$

- **Error-state vector**, where δp_F are the stacked errors of all active feet:

$$\delta x = \begin{bmatrix} \delta\theta & \delta v & \delta p & \delta p_F \end{bmatrix}^T$$

- **Process Jacobian** $F_k = \frac{\partial f}{\partial x}\big|_{\hat{x}_{k-1}}$ used to propagate the error state $\delta x_k \approx F_k \delta x_{k-1}$:

$$F_k = \begin{bmatrix} F_{\theta\theta} & 0 & 0 & 0 \\ F_{v\theta} & I & 0 & 0 \\ F_{p\theta} & F_{pv} & I & 0 \\ 0 & 0 & 0 & I \end{bmatrix}$$

$$F_{\theta\theta} = \Delta R^T \qquad F_{v\theta} = -R_{WB,k}[a_B]_\times \Delta t$$

$$F_{p\theta} = -\frac{1}{2} R_{WB,k}[a_B]_\times \Delta t^2 \qquad F_{pv} = I\,\Delta t$$

- Process noise Q captures model uncertainty:

$$Q = \mathrm{diag}(Q_\theta,\ Q_v,\ Q_p)$$

Contact Update: Measurement Model

- For a stance foot, predict the foot position from the current base estimate:

$$\hat{p}_{F,W} = p_W + R_{WB} p_{F,B}$$

- Compare this with the stored foot anchor:

$$r = \hat{p}_{F,W} - p_{F,W}$$

- If the foot is really fixed, this residual should be close to zero.
- The EKF update uses this residual to correct:
  - base orientation,
  - base position,
  - foot anchor position.

Key assumption

The stance foot does not slip.

Contact Update: Measurement Jacobian

- The residual is:

$$r = p_W + R_{WB} p_{F,B} - p_{F,W}$$

- **Measurement Jacobian** $H_k = \frac{\partial h}{\partial x}\big|_{\hat{x}_k^-}$ maps the error state to the residual $r_k \approx H_k \delta x_k$:

$$H = \begin{bmatrix} H_\theta & H_v & H_p & H_{p_F} \end{bmatrix}$$

- Main blocks:

$$H_\theta = -R_{WB}[p_{F,B}]_\times, \quad H_v = 0, \quad H_p = I, \quad H_{p_F} = -I$$

- *Remark:* The orientation Jacobian H_θ can be unreliable in real world scenarios due to foot slip, imperfect contact detection, kinematic modeling errors, compliance and soft ground, frame misalignment, etc.
- For robustness, set $H_\theta = 0$ on real data. Position terms still constrain the base position:

$$H_p = I, \quad H_{p_F} = -I$$

Contact Update: Kalman Correction

- Innovation covariance:

$$S = HP^{-}H^{T} + R$$

- Kalman gain:

$$K = P^{-}H^{T}S^{-1}$$

- Error-state correction:

$$\delta x = -Kr$$

- Inject the correction:

$$R_{WB} \leftarrow R_{WB}\exp([\delta\theta]_{\times})$$

$$v_W \leftarrow v_W + \delta v$$

$$p_W \leftarrow p_W + \delta p$$

$$p_{F,W} \leftarrow p_{F,W} + \delta p_F$$

Contact Update: Covariance

- Use the Joseph-form covariance update:

$$P = (I - KH)P^{-}(I - KH)^{T} + KRK^{T}$$

- This is more numerically stable than:

$$P = (I - KH)P^{-}$$

- It helps preserve:

$$P = P^{T}, \qquad P \succeq 0$$

- In code, symmetrize after the update:

$$P \leftarrow \frac{1}{2}(P + P^{T})$$

What Can Go Wrong?

- **IMU noise**: causes random drift after integration.
- **IMU bias**: causes systematic drift, often much worse than noise.
- **No contact**: the robot is temporarily dead-reckoning from IMU only.
- **Foot slip**: violates the fixed-contact assumption.
- **Wrong noise tuning**:
  - small Q: filter trusts prediction too much
  - large R: filter ignores foot updates
  - small R: filter over-trusts possibly slipping contacts

Tutorial Workflow

1. Run the provided simulation and inspect the data.
2. Visualize the EKF state and active foot states.
3. Implement IMU prediction.
4. Test IMU-only estimation.
5. Add IMU noise and bias.
6. Implement foot-contact correction.
7. Add foot slip and observe failure cases.
8. Tune Q and R.
9. Visualize covariance ellipses.
10. Apply the same estimator to real quadruped data.

Task 1–2: Data and State Inspection

- Start with synthetic data. This provides ground truth for evaluation as follows:

$$e_p(t) = p_{\text{est}}(t) - p_{\text{gt}}(t)$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} \|e_p(k)\|^2}$$

- Inspect the initialized state:

$$x = [R_{WB}, v_W, p_W]$$

- Then add stance-foot states using contact information:

$$x = [R_{WB}, v_W, p_W, p_{F,W}^{(1)}, \ldots]$$

Task 3: Implement IMU Propagation

1. Propagate orientation:
$$R_k = R_{k-1} \exp([\omega_k]_\times \Delta t)$$
2. Compute world-frame acceleration:
$$a_W = R_k a_B + g_W$$
3. Integrate velocity:
$$v_k = v_{k-1} + a_W \Delta t$$
4. Integrate position:
$$p_k = p_{k-1} + v_{k-1}\Delta t + \frac{1}{2} a_W \Delta t^2$$
5. Build F_k and propagate:
$$P_k^- = F_k P_{k-1} F_k^T + Q$$

Task 4–5: IMU Noise and Bias

- Add random IMU noise (zero-mean Gaussian noise w_a, w_ω):

$$a_m = a + w_a, \qquad \omega_m = \omega + w_\omega$$

- Add systematic IMU bias (b_a, b_ω):

$$a_m = a + b_a + w_a$$

$$\omega_m = \omega + b_\omega + w_\omega$$

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$$a_m = a + b_a + w_a$$

$$\omega_m = \omega + b_\omega + w_\omega$$

Answers

- Noise causes drift because acceleration and angular velocity are integrated.
- Bias causes larger long-term drift due to accumulation over time.
- Gyro bias is especially harmful:

$$\omega_m = \omega + b_\omega \Rightarrow R_{WB} \text{ drifts over time} \Rightarrow a_W \text{ is rotated incorrectly} \Rightarrow p_W \text{ drifts}$$

Task 6: Implement Foot-Contact Correction

1. Compute predicted foot position:

$$\hat{p}_{F,W} = p_W + R_{WB} p_{F,B}$$

2. Compute residual:

$$r = \hat{p}_{F,W} - p_{F,W}$$

3. Build measurement Jacobian:

$$H = \begin{bmatrix} -R_{WB}[p_{F,B}]_\times & 0 & I & -I \end{bmatrix}$$

4. Compute:

$$S = HP^- H^T + R, \qquad K = P^- H^T S^{-1}$$

5. Correct state:

$$\delta x = -Kr$$

6. Update covariance using Joseph form.

Task 7: Foot Slip

- Introduce foot slip in the simulation.

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  Answer
- The contact update assumes $p_{F,W} \approx$ constant.
- Foot slip violates this assumption:

$$p_{F,W}(t + \Delta t) \neq p_{F,W}(t)$$

- Then the residual is no longer caused only by state-estimation error.
- The EKF may apply a wrong correction:

$$\delta x = -Kr$$

- Result:
  - position error can increase,
  - orientation can be pulled in the wrong direction,
  - covariance can become overconfident,
  - the estimate can become inconsistent.

Task 8: Tuning Q and R

- Process noise Q controls trust in prediction:

$$P^- = FPF^T + Q$$

- Measurement noise R controls trust in contact updates:

$$S = HP^-H^T + R$$

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- Process noise Q controls trust in prediction:

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Answer

- If Q is too small:
  - the filter trusts the IMU prediction too much,
  - covariance becomes too small,
  - contact corrections have too little effect.
- If R is too large:
  - the filter ignores foot contacts,
  - behavior approaches IMU-only dead reckoning.
- If R is too small:
  - the filter over-trusts contacts,
  - foot slip can strongly corrupt the estimate.

Task 9: Covariance Ellipses

- The position covariance in the x-y plane is:

$$P_{xy} = \begin{bmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{bmatrix}$$

- Covariance ellipses visualize uncertainty.

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- The position covariance in the x-y plane is:

$$P_{xy} = \begin{bmatrix} P_{xx} & P_{xy} \\ P_{yx} & P_{yy} \end{bmatrix}$$

- Covariance ellipses visualize uncertainty.
  Answer
- Ellipse size $\sim$ uncertainty (larger = higher, smaller = lower)
- IMU-only prediction - P grows; Contact updates - P shrinks.
- During flight phases, no foot anchor is available, so uncertainty grows.

Task 10: Real Data

- Apply the same EKF for real world data and tune the Q and R noise parameters.

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- Apply the same EKF for real world data and tune the Q and R noise parameters.
  Answer
- Real data is harder than simulation because of:
  - IMU bias and calibration errors,
  - time synchronization errors,
  - imperfect contact detection,
  - compliance and impacts,
  - kinematic errors,
  - foot slip,
  - uneven terrain.
- For rough terrain, possible improvements include:
  - contact probability estimation,
  - slip detection,
  - IMU bias estimation,
  - adaptive measurement noise,
  - visual or LiDAR odometry fusion.

Key Takeaways

- EKF combines:
  - IMU-based prediction (drift-prone),
  - contact-based corrections (drift-reducing).
- Stance feet act as temporary world-frame anchors.
- Performance depends strongly on:
  - noise tuning (Q, R),
  - contact quality (slip, detection),
  - modeling assumptions.
- Real-world data is more challenging than simulation.
- Error-state formulation enables efficient handling of orientation in $\mathrm{SO}(3)$.